

DEPARTMENT OF PHYSICS UNIVERSITY OF PERADENIYA

Guidelines for Lab Report Writing

Format of the Laboratory Report

Main parts of the report:

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| <ul style="list-style-type: none"> (a) Cover Page (b) Title (c) Aim(s) <li style="padding-left: 20px;">(d) Materials and Apparatus <li style="padding-left: 20px;">(e) Theory <li style="padding-left: 20px;">(f) Procedure <li style="padding-left: 20px;">(g) Observations and Calculations <li style="padding-left: 20px;">(h) Error Estimation (i) Results (j) Discussion (k) References (l) Observation Sheet | <div style="display: inline-block; vertical-align: middle; font-size: 3em;">}</div> <div style="display: inline-block; vertical-align: middle;"> <p>If there are multiple aims and if it is appropriate, (d) to (h) can be repeated in the same order.</p> </div> |
|--|---|

Section	Description
Cover Page	Include the course code, the experiment number, the title of the experiment, the name and the registration number of the student, the group number, the name and the registration number of the laboratory partner(s), name of the demonstrator, and the date of the experiment conducted.
<u>Title</u>	The title of the experiment.
<u>Aim</u>	The purpose of the experiment.
<u>Materials and Apparatus</u>	A list of all the equipment, items and materials used in the experiment.
<u>Theory</u>	The theory involved in the experiment (including assumptions, essential equations, figures, etc.) Equations that are referred later on should be numbered using Arabic numbers within brackets toward right side of the page.
<u>Procedure</u>	It is written in <i>past passive voice</i> . A sufficient description of the experimental and data collection procedures. Include a brief explanation about the <u>data analysis</u> . Include diagram(s) (set up, circuit diagrams, etc.) if necessary.
<u>Observations and Calculations</u>	The readings obtained (and brief explanation of observations, <u>if any</u>), relevant calculations (including data analysis, comparisons, verifications, etc.), tables and graphs should be included wherever necessary.
<u>Error Estimation</u>	A quantitative estimation about the uncertainty of the final results of the experiment. Include an explanation on how errors are propagated to the final results.
<u>Results</u>	The final results of the experiment with the uncertainties.
<u>Discussion</u>	In the discussion, you may compare your results with known reported values and explain reasons for any deviations. You may

	also discuss the limitations of the experiment and how the experiment can be improved to obtain better results. Limit the discussion to about 100 words.
<u>References</u>	A list of references (if any).
<u>Observation Sheet</u>	Include the date, experiment number, registration numbers and all observations and measurements (in ink) taken during the experiment. Should be signed by the demonstrator before leaving the lab class.

All titles should be written in "title case", that is, capitalize first letter of each word except for prepossessions. All subtitles should be underlined.

Guidelines for drawing borders

When the report is either handwritten or printed on A4 sheets, a boarder should be drawn around each sheet of paper. This boarder margins should have following widths.

- ❖ Left Margin = 2 cm
- ❖ Other three margins = 1 cm

Guidelines for drawing figures

- ❖ Figure caption (eg: Fig. 1. Caption) must be written at the bottom of the figure
- ❖ Figure must be drawn with a pencil to an approximate scale
- ❖ Apparatus must be labeled with horizontal (vertical) arrows
- ❖ Figure should be center aligned
- ❖ Figure should appear after (before when written in a record book) it has been referred in the text

Guidelines for tables

- ❖ Table caption (eg: Table 1. Caption) must appear above the table
- ❖ Table border and lines must be drawn in pencil
- ❖ Appropriate SI units should be written with the column/row title
- ❖ Table should be center aligned
- ❖ Data values should be center aligned within the columns

Guidelines for drawing graphs

- ❖ Title of the graph (eg: Graph 1: y vs x of or The graph of y vs x of) should be (written in ink) at the top of the graph and underlined.
- ❖ Axes should be drawn, scaled and labeled with pencil.
- ❖ Scale should be both readable and rational. A major division line of the graph sheet should always represent a decimal multiple of 1, 2, 4, or 5. ((e.g. 1, 2, 0.4, 0.05, 20, 500, etc.).
- ❖ The maximum possible space of the graph sheet should be utilized given that the scales are readable and rational.
- ❖ Data points should be marked and lines or curves should be drawn with a pencil.
- ❖ Include the error bars when necessary. (Use a pencil)

Note: The best-fit line may not go through any of the data points.

➤ *Example of a non-readable scale and irrational*

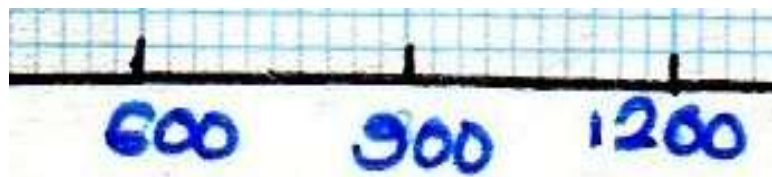


30 minor divisions = 10 units

1 minor division = $(10/30)$ units = 0.3333..... units

Scale is non-readable

➤ *Example of a readable, irrational scale*



10 minor divisions = 300 units

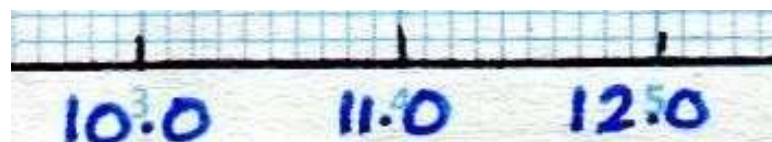
1 minor division = $(300/10)$ units = 30

The scale is readable.

But $(1/\text{unit represented by minor division}) = (1/30) = 0.0033333.....$

The scale is irrational.

➤ *Examples of readable, rational scales.*



Example 01

10 minor divisions = 1 unit

1 minor division = $(1/10) = 0.1$ units.

$(1/0.1) = 10$

The scale is readable and rational.

Example 02



10 minor divisions = 0.2 units

1 minor division = $(0.2/10) = 0.02$

$(1/0.02) = 50$

The scale is readable and rational.

Guidelines for writing results

- ❖ Final results should be written using scientific notation in SI/relevant units with the correct significant figures.
For example, the result (91.2 ± 1.2) mm should be written as $(9.12 \pm 0.12) \times 10^{-2}$ m.
- ❖ Standard/reported values of the parameter determined should be written if they are available (include the source as a reference).
- ❖ If the final results depend on a specific condition (temperature, pressure for instance), you may state the condition as well.

Guidelines for writing the discussion

In the discussion, consider the following;

- ❖ The standard/reported value of the quantity you determined.
- ❖ Compare your value with the standard/reported value and discuss possible reasons for any deviations.
- ❖ Ways of minimizing the error/s.
- ❖ Suggestions for improvement of the experiment to increase the accuracy.
- ❖ Comment on standard and more accurate method(s) to determine the same parameter (with reference).
- ❖ Any other relevant matter(s).

Guidelines for references

Following are the formats for a few different types of references.

❖ *Cite a website in the following format.*

- i). Author(s) (initials followed by the surname) or the organization responsible for the site
- ii). Year or date created or last updated (in brackets)
- iii). Page title (in italics)
- iv). Publisher of site (if available)
- v). URL or internet address
- vi). Retrieved date

Example:

C.R. Nave (2016) *Hyper Physics*. Department of Physics and Astronomy, Georgia State University.
Retrieved from <http://hyperphysics.phy-astr.gsu.edu/hbase/hframe.html> on 14th January 2018.

❖ *Cite a book in the following format.*

- i). Author(s) initial(s) and surname(s)
- ii). Title of the book (in italics)
- iii). Publisher, City, Year (in brackets)
- iv). Volume, Chapter, Section, Page etc.

Example:

J. Walker, D. Halliday, R. Resnick, *Fundamentals of Physics* (Wiley, New Jersey, 2011), Vol. 1, Chap. 14, p. 362

General Guidelines

- ✓ Read the handouts and come prepared to the lab.
- ✓ Reports must be hand written including graphs. (**NO Printouts**)
- ✓ Use A4 sheet (write only on one side)
- ✓ Graphs and figures must be attached on the same side as the rest of the report. (right side)
- ✓ All headings and paragraphs should begin at the margin.
- ✓ Best fit for $y = mx$ type graph does not necessary to go through (0,0). The line should be drawn symmetrically in-between data points and through the average point ($\bar{x}$, $\bar{y}$).
- ✓ Ignore any bad points when drawing the best fitting line. You may give an explanation in the discussion about these points.
- ✓ Draw rough graphs and sketches during the lab to check for bad data points.
- ✓ The observation sheet signed by the demonstrator should be attached to the report.
- ✓ Lab report must be submitted on or before the dead line. (**Late submission will not be graded**)

Examples of accurately drawn lines of best fit

